Shift Register Display

Send one bit at a time; reveal one stable digit

The displayed glyph and three named test points explain the transfer without Serial.

Lesson	010 — component	**Time**	120–180 minutes
Prerequisites	Lessons 001–009	**Board**	Arduino Mega 2560
Evidence	glyph, TP-DATA, TP-CLOCK, TP-LATCH	**Status**	host verified; bench open
Safety	E1: USB-powered logic and resistor-limited LED segments only		

1 Mission and evidence chain

Build a decimal counter from two high-level objects: `ShiftRegisterOutput` owns the three control pins, while `SevenSegmentDisplay` turns a glyph into the register's eight output bits. The sketch reads in the same order as the circuit:

$$\text{time} \rightarrow \text{choose digit} \rightarrow \text{serialize bits} \rightarrow \text{latch once} \rightarrow \text{stable glyph}.$$

The D13 diagnostic LED proves that both components acquired their pins. The display proves the current glyph. TP-DATA, TP-CLOCK, and TP-LATCH expose the electrical transfer to a logic probe or oscilloscope. These observations prove different claims; none alone proves the entire chain.

1.1 Learning outcomes

You will be able to:

- distinguish a shift clock from the output latch;
- predict the register byte for a decimal glyph;
- explain why the visible display remains stable while new bits shift;
- use deterministic RAII ownership and a nonblocking update loop;
- separate acquisition, behavior, and safe-state evidence.

> **Safety checkpoint.** Identify the exact 74HC595 and display part numbers before wiring. Seven-segment package pinouts vary. A common-anode display wired as common-cathode gives misleading results and can exceed an LED rating. Remove USB power before moving wires or meter clips.

2 Parts and current budget

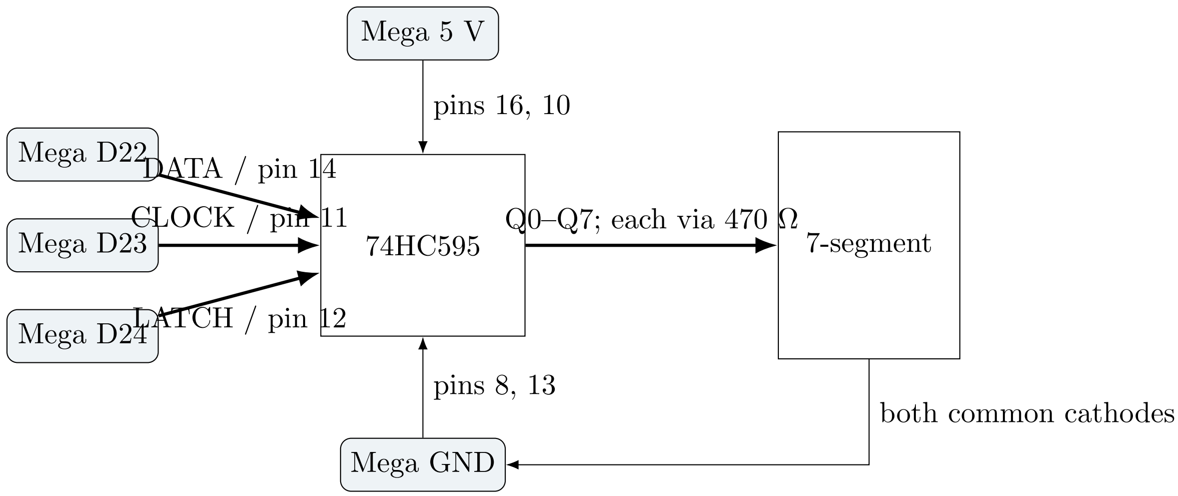
Quantity	Part	Requirement
1	Arduino Mega 2560	USB-powered reference board
1	74HC595	5 V HC-family eight-bit serial-in/parallel-out register
1	Seven-segment display	One digit, common cathode, known pinout
8	470 Ω resistors	One per segment, including decimal point
1	Breadboard and jumpers	Short, inspectable connections
1	Logic probe or oscilloscope	Optional; multimeter still verifies static levels

For each lit segment, calculate

$$I_{\text{segment}} = \frac{V_{\text{CC}} - V_F}{470\ \Omega}.$$

Record the actual LED forward-voltage specification. The lesson limit is 8 mA per segment and 40 mA total display current; increase the resistor value if either calculation exceeds its limit. These conservative lesson limits do not replace the register and display datasheets.

3 Authoritative wiring

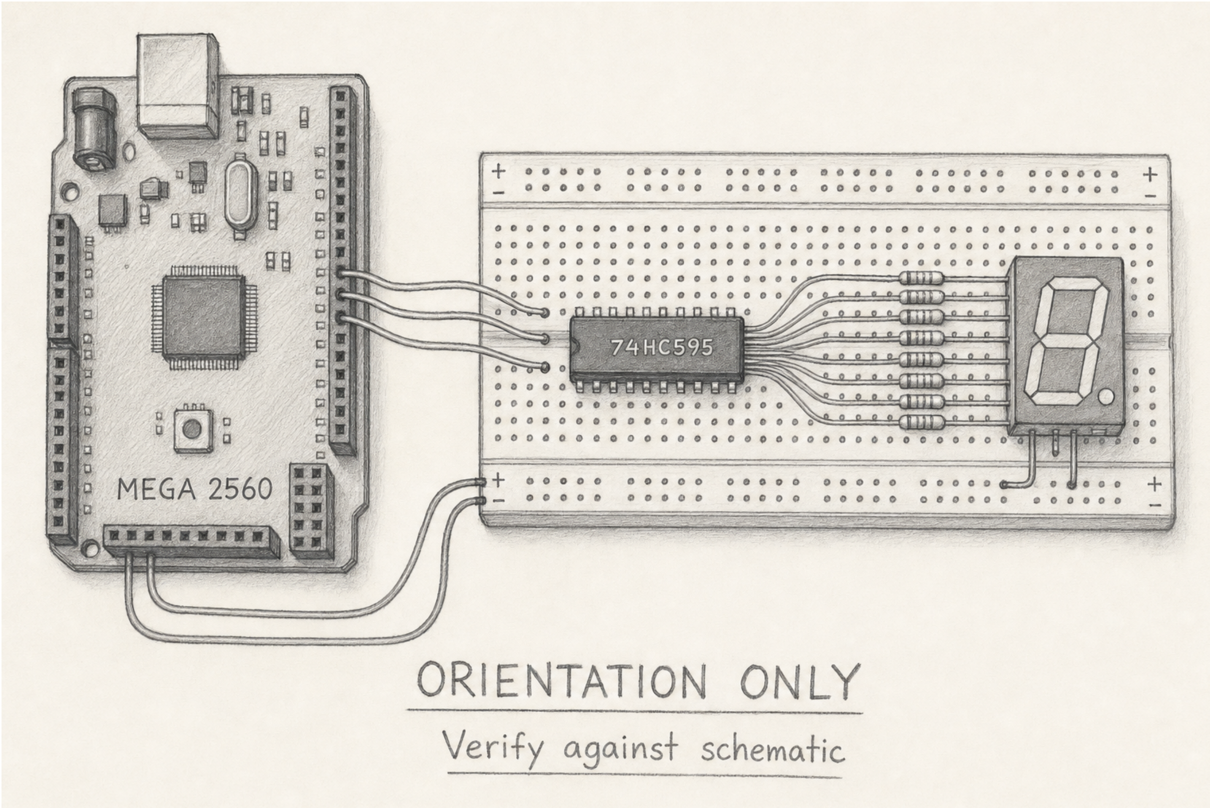


Pin-by-pin connection list

- D22 / TP-DATA → DS, 74HC595 pin 14.
- D23 / TP-CLOCK → SHCP, pin 11.
- D24 / TP-LATCH → STCP, pin 12.
- 5 V → VCC pin 16 and active-low clear pin 10.
- GND → GND pin 8 and active-low output-enable pin 13.
- Q0 through Q6 → segments a through g; Q7 → decimal point. Every path has its own 470 Ω resistor.
- Both display common-cathode pins → GND. Use the display datasheet to locate them.

Place a 100 nF ceramic bypass capacitor directly between pins 16 and 8. The pencil plate helps with physical orientation only. In this introductory three-wire circuit, active-low OE is tied to GND, so the output drivers are enabled before the sketch runs. The 74HC595 storage register's power-up state is not specified; reset or power-up can therefore show an arbitrary, possibly bright, segment pattern until the first blank or glyph is latched. The software lifecycle cannot prevent that startup transient.

A later hardware upgrade gives OE its own Mega output and a pull-up resistor. The pull-up keeps the register outputs disabled while the Mega resets; software then shifts and latches the blank value before driving OE low. That fourth control signal is required when blank output during power-up is part of the circuit contract.



Alternative text: a Mega, breadboard, 74HC595, eight resistors, and one-digit display arranged left to right; the exact list above is authoritative.

4 Interface and ownership

```
constexpr adk::ShiftRegisterPins displayPins = {22, 23, 24};

adk::Runtime             runtime;
adk::SevenSegmentDisplay display (
    runtime.resources (),
    displayPins,
    adk::SevenSegmentPolarity::CommonCathode);

display.initialize ();
display.show (adk::SevenSegmentGlyph::Five);
display.blank ();
display.shutdown ();
```

`SevenSegmentDisplay` composes one `ShiftRegisterOutput`. Initialization claims D22, D23, and D24 transactionally. A conflict leaves no partial claim. Repeated initialization succeeds without another transfer. `show()` accepts digits, hexadecimal glyphs, dash, blank, and an optional decimal point.

For common cathode, a one bit lights its segment. For common anode the API inverts all eight output bits. The wiring and constructor polarity must agree. The supported mapping is Q0–Q6 to a–g and Q7 to the decimal point.

`shutdown()` first shifts the polarity-correct blank value, then makes DATA, CLOCK, and LATCH high impedance and releases their claims. Destruction does the same during normal scope exit or exception unwinding in a caller. ADK itself neither throws nor allocates.

4.1 A transfer without update flicker

For each byte, `ShiftRegisterOutput::show()` holds LATCH low, presents the most-significant remaining bit on DATA, and pulses CLOCK. After eight clock pulses it pulses LATCH once. The output register changes as one visible step, so intermediate shifted bytes are not shown during normal updates. This does not control the register's unspecified state before the first software latch while OE remains tied low.

> **Predict:** glyph 8 uses segments a–g, so its common-cathode value is `0x7f`. **Observe:** seven segments lit and decimal point dark. **Interpret:** the stable output matches the glyph encoding. It does not prove clock timing margins; TP-CLOCK and TP-LATCH supply that evidence.

5 Narrative run loop

The canonical sketch declares the runtime, display, and diagnostic LED in dependency order. `setup()` reads "acquire circuit, show ready." `loop()` reads "observe time, decide whether the count is due, choose the digit, show the digit." No `delay()` blocks observation.

```
void loop ()
{
    if (!ready)
    {
        return;
    }

    const adk::TimePoint now (millis ());
```

```
    if (!countIsDue (now))
    {
        return;
    }

    const adk::SevenSegmentGlyph digit = chooseDigit ();
    showDigit (digit);
}
```

The complete, compile-authoritative sketch is the Lesson 010 sketch download. Build and upload it only through repository targets; no particular IDE is required.

6 CLI experiment

```
make bootstrap
make arduino-Lesson010ShiftRegisterDisplay
make upload-Lesson010ShiftRegisterDisplay PORT=/dev/ttyACM0
make observe-new LESSON=010
make observe-check CARD=hardware/acceptance/lesson-010.md
```

Serial is optional supporting evidence:

```
make monitor PORT=/dev/ttyACM0 BAUD=115200
make serial-log PORT=/dev/ttyACM0 BAUD=115200
```

The canonical sketch does not require or emit Serial messages. A missing serial log therefore says nothing about display correctness.

6.1 Predict, observe, interpret

Place	Prediction	Interpretation
D13	Steady on after setup	All four output pins were acquired; not proof of segment wiring
Display	0, then one increment each second	Glyph selection and latched outputs agree visibly
TP-DATA	Eight data levels per update	Serialized value is present; probe with GND reference
TP-CLOCK	Exactly eight pulses per update	Eight bits were shifted
TP-LATCH	One pulse after the eight clocks	The visible output changes atomically
D22–D24 after shutdown	High impedance	Endpoint safe state; measure rather than infer from blank display

6.2 Experiment sequence

1. With USB removed, inspect every IC pin, resistor, display common, and the bypass capacitor against the exact part datasheets.
2. Predict the first three glyphs and the CLOCK:LATCH pulse count.
3. Attach probes with power removed. Reference every measurement to Mega GND. Apply USB power only after a second inspection.
4. Confirm D13 first, then the visible 0–9 sequence.
5. Observe TP-DATA, TP-CLOCK, and TP-LATCH. Save one annotated capture.
6. Remove USB power. Do not move bare probes around the live breadboard.

Host-verified is the current release status. Do not fill the measured columns or mark hardware accepted until the actual Mega circuit has been observed and recorded.

7 Diagnosis by evidence

Observation	Distinguishes	Next check
D13 dark	acquisition failed or setup stopped	pin conflicts, exact status in host test, then D13 circuit
D13 on; display dark	control acquired but display path unknown	5 V/GND, OE low, clear high, display polarity
Wrong stable glyph	output path exists; mapping differs	Q0–Q7 order and segment a–dp datasheet pins
Partial glyph	one or more segment paths absent	resistor, LED segment, Q output continuity unpowered
Flicker or mixed glyph	latch behavior or supply integrity suspect	TP-LATCH pulse count and 100 nF bypass
No CLOCK pulses	transfer did not occur	TP-DATA/TP-CLOCK, initialization, pin conflict
Eight clocks; no latch	shift register changed internally only	TP-LATCH wiring and pin 12
Chip or resistor hot	unsafe current or wiring	remove USB immediately; recheck identity and current budget

7.1 Fault injections without rewiring live

- In a host test, pre-claim CLOCK. Expect `ResourceBusy`, no partial DATA/LATCH ownership, and no output transfer.
- Construct an object with duplicate data and clock pins. Expect `InvalidArgument` before a claim or hardware write.
- Request every glyph twice and compare the recording fake's traces. They must match byte for byte.
- Advance timestamps across 32-bit wrap in the example timing model. The elapsed-time decision must remain correct.

8 Design exercises

1. Write the common-cathode bytes for 0, 1, 2, and dash. Then invert each byte for common anode.
2. Why is one resistor on the display common pin not equivalent to one resistor per segment?
3. Extend the counter to hexadecimal without changing `ShiftRegisterOutput`.
4. Add a decimal-point heartbeat that does not block the counter. Name the independent state and deadline.
5. Sketch two cascaded 74HC595 registers. Which signal is shared, which output feeds the next DATA input, and which resource remains owned?
6. Explain what a stable digit proves that a TP-CLOCK capture does not, and vice versa.

9 Bench acceptance record

Date	__________	Reviewer	__________
Board	Mega 2560 revision: __________	USB supply	__________
74HC595 marking	__________	Display marking	__________
Display polarity	__________	Resistors	__________
Measured 5 V	__________	Segment V_F	__________
Segment current	__________	Maximum total	__________

9.1 Required observations

Check	Pass	Recorded evidence
Unpowered wiring matches both datasheets	☐	
D13 separately proves successful acquisition	☐	
Digits 0–9 are stable and correctly mapped	☐	
TP-CLOCK shows eight pulses per update	☐	
TP-LATCH shows one later pulse per update	☐	
Startup transient with OE tied low is observed and recorded	☐	
First software latch establishes the documented zero	☐	
Reset returns to the documented initial sequence	☐	
Shutdown blanks, then D22–D24 become high impedance	☐	
Power removal leaves every part cool and dark	☐	

Instrument and capture identifiers: ____________________

Deviations or open questions:

Stop conditions: unexpected heat, unstable USB, a display current above the recorded budget, signal outside the rails, unexpected simultaneous segments, or disagreement between the actual part datasheet and this lesson. Remove USB power physically; software shutdown is not the emergency stop.

Deferred upgrade: repeat the startup and reset observations after adding a pulled-up, software-owned OE signal. Record that outputs remain disabled until a blank byte has been shifted and latched; do not transfer that claim to the three-wire circuit above.

10 What comes next

Lesson 011 keeps this display visible while a deterministic, nonblocking traffic state advances and records pedestrian requests. Lesson 012 composes the display, buttons, and mutually exclusive signal lamps into the next three-lesson project.